GOLDENWAY — SOLUTION ARCHITECTURE SUMMARY

Cashless top-ups for Golden Arrow Bus Service (GABS) commuters

Group 26 • Class 3D – PFP
Adams – Barends – Barron – Everts

1. INTRODUCTION & SCOPE

A digital ticketing & Gold Card top-up platform for Cape Town's public transport commuters — replacing cash only ticket booths with instant, cashless recharges.

THE PROBLEM
Commuters lose real time and money to long queues, cash only booths, and booths that are frequently closed during peak hours.

THE SOLUTION
A mobile/web platform for instant Gold Card top-ups via card or EFT — no queue, no kiosk visit required.

THE OUTCOME
A 20-minute queue becomes a seconds long transaction — cashless, safer, and available 24/7 from any device.

2. HIGH-LEVEL ARCHITECTURE

Client side app → application logic → data layer → external services, deployed as a static/serverless web app.

BROWSER / MOBILE (CLIENT)
- React single-page app
- Responsive UI, mobile-first
- Login, Home, Load Trips, Card, History, Profile

APPLICATION LAYER
- Supabase Auth (login/signup)
- Top-up & balance business logic
- API calls to external services

DATA LAYER
- Supabase (Postgres)
- Users, transactions, ride/top-up history

EXTERNAL SERVICES
- Stripe / PayFast (payments)
- GABS Gold Card system (future API)

 GitHub repo Vercel build Live production URL Shareable, HTTPS everywhere

3. COMPONENT BREAKDOWN

Screens / modules that make up the app.

Splash Screen | Login Screen | Home Screen | Load Trips | Card / Gold Pass | Ride History | Support | Profile

5. TECHNOLOGY STACK

React Frontend UI	**Supabase** Backend, auth, DB	**Stripe/PayFast** Payments
Vercel Hosting	**GitHub** Version control	**Figma** UI prototyping

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4. KEY DATA FLOWS

SIGN-UP / LOGIN

Open app → Enter credentials → Supabase Auth verifies → Session created

TOP-UP

Select amount → Stripe/PayFast processes → Transaction recorded

Gold Card balance updates → Confirmation shown

6. SECURITY CONSIDERATIONS

CURRENT

- HTTPS/SSL on all traffic
- Supabase managed authentication
- PCI DSS compliance via PayFast – card data never touches our servers

FUTURE

- Rate limiting on auth & payment endpoints
- Fraud detection on top-up transactions
- Audit logging for financial actions

7. FUTURE ARCHITECTURE PROPOSAL

- GABS Gold Card API integration for real time balance sync
- Native mobile app (iOS/Android) alongside the web platform
- ⚙ Offline top-up queuing for low connectivity areas